

**Proposal for an Individualized Instruction of
Process Synchronization Represented on Low-Cost Hardware**

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Instructor: Dr. Daniel Bennett

Study to be completed: Fall 2026

I. Area of Study

This independent study will allow me to develop a deeper understanding of process synchronization. The goal of this study is to investigate, understand, and implement process synchronization on a lower-cost scale. The hardware would consist of a cluster of microcontrollers, which are to be wired onto bread boards, and a computer to accompany by providing a user interface and access. The languages employed are C, C++, and an embedded version of C used by Arduino boards (colloquial name "Arduino language"). This study stems from Dr. Dan Bennett's Spring 2026 class of Operating Systems, where there he introduced the students to microprocessors and how they can be used to represent many problems present in computing, including process synchronization.

Languages

C

C is a general-purpose programming language developed by Dennis Ritchie at Bell Labs in the early 1970s, originally created to reimplement the Unix operating system and intended for systems programmers. It's known for being able to directly access hardware and memory, which system programmers wanted while they were working on operating systems like Unix. One case of using C is a programmer being presented with a single flat address space in which variables correspond closely to physical memory locations and may be manipulated directly. This can be done with pointers, giving control over how data is stored, addressed, and modified.

C++

C++ is a general-purpose programming language created by Bjarne Stroustrup that was originally developed as an extension of C, adding object-oriented features. It was originally designed with system programming, resource-constrained software, and large systems in mind. It highlights performance, efficiency, and flexibility of use. It has additions to facilitate low-level memory manipulation for systems like microprocessors or to make operating systems.

Arduino Language

The Arduino Language is the programming syntax used within ArduinoIDE for working with microprocessors and microcontrollers. "Arduino Language" in itself is not a language, but a subset of C++ optimized for the very low memory of microprocessors. It highlights efficiency, as well as very low memory usage. Understanding C++ is enough to understand this syntax.

Hardware

Microprocessors

A microprocessor is a computer processor that contains the arithmetic, logic, and control circuitry required to perform the functions of a computer's Central Processing Unit (CPU). They

are capable of running machine code and performing arithmetic operations. These are similar to fully functional commercial computers in operation and programming.

Distributed embedded system

A distributed embedded system is a set of computer nodes that are interconnected by a communication network or protocol. This can be accomplished via normal network connections. A distributed embedded system's processing elements may include a normal computer or a microcontroller. Each microcontroller acts independently within the system and would be running its own program, as well as possessing its own local memory.

II. Description

For this independent study, I will examine the topic of distributed process synchronization with the attempt of representing the idea on low-cost hardware. The goal is to understand process synchronization at a lower level, while also examining and navigating the limitations of a low-cost hardware representation. During this, I'll utilize the computer labs provided by PennWest Edinboro's Computer Science Department for working on this project. Before beginning the study, I will research what it takes to make a distributed embedded system by investigating classical computer science problems such as the Cigarette Smokers problem, and the Dining Philosophers problem.

This study will allow me as a student the opportunity to dive deeper into microprocessors on a lower level. This will also allow me to explore the advantages and challenges of replicating classic problems on low-cost hardware. Testing parameters will be much more critical on low-cost hardware when compared to a regular desktop computer. Because of this, optimization in programs I run will need to be a top priority while implementing. Microprocessors have very limited memory, especially for programs. Memory sizes are a fraction of a normal desktop computer. On top of all of that, accomplishing this will show that expensive equipment isn't required to explore complex computer science topics involved with classic problems.

III. Plan of Study

1. Setup environment

Time to complete: 2 weeks

- a. Determine microprocessor boards to use, as well as display output, i.e. ESP32, MEGA328P to monitor via USB-to-Serial UART.

2. Single processor implementation

Time to complete: 3 weeks

- a. Investigating classic computer science problems to represent on single processor hardware.

3. Integrating network programs

Time to complete: 2 weeks

- a. Implementation of communication programs for multiprocessor systems.

4. Distributed process synchronization implementation

Time to complete: 4 weeks

5. Testing/Evaluation

Time to complete: 3 weeks

- a. Gathering data from project to complete and submit promised deliverables.
Course evaluation from instructor.

E-1. Experimenting with other hardware

- a. Potentially implementing project idea on 8-bit hardware or other limited hardware.

E-2. Other problems

- a. Exploring implementation of more classic computer science problems.

IV. Deliverables

1. Report on Study
 - a. 4-6 paged report on study that consists of:
 - i. Formal research conducted
 - ii. Annotated bibliography
2. Presentation of Studies
 - a. Formal presentation to PennWest Edinboro Department of Computing and Engineering Technology faculty and students
 - i. This could include a demonstration of the systems assembled as well as a chance to ask questions about the entire process
3. Submitting Project to The Pennsylvania Association of Computer and Information Science Educators (PACISE) 2027
 - a. Birds-of-a-feather session
 - b. Paired with non-reviewed paper/report of project

V. Special Topics in Computer Science – CMAC 3990

With this individualized instruction, I intend to fulfill the requirements for the Special Topics in Computer Science – CMAC 3990. This independent study will introduce me to the principles of computer science research. A strong focus on researching and presenting older problems i.e. cigarette smokers, dining philosophers, producer-consumer will be integrated into the final report of the research conducted.

The following will be met:

- Research paper
- Annotated bibliography
- Presentation of research conducted

Works Cited

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